

SAI SRINIVAS TATWIK MEESALA

tatwik.m.19@gmail.com |  Sai Srinivas Tatwik Meesala |  Tatwik19 |  +1 (602) 815-2130

SUMMARY

Passionate Robotacist with expertise in embedded systems, perception, industrial automation, autonomous vehicles, & aerial robotics. Demonstrated leadership in engineering projects, excelling in documentation, CAD design, embedded components, digital twin, & programming. Inspired to reach the pinnacle in robotics with a meticulous approach.

EXPERIENCE

Arizona State University – Tempe/Mesa, Arizona

Robotics & Applied Linear Algebra Teaching Assistant

Jan 2023 – May 2024 \ Jan 2025 - Present

- Instructed robotic systems design, covering kinematics, dynamics, modeling, and control.
- Designed industrial-focused labs integrating environment- and context-aware robotics using LLMs, including training on FANUC and UR5e robots, digital twin workflows, and UAV platforms.
- Assisted in graduate-level linear algebra for engineering.

Arizona State University – TSMC P-SMART Partnership – Tempe/Phoenix, Arizona

Teaching Assistant, SACA C-101/C-102 Industry 4.0 Training

May 2026

- Taught SACA C-101/C-102 smart automation modules covering Industry 4.0 fundamentals, sensors, pneumatics, electrical control, PLC/HMI, robotics terminology, and semiconductor equipment practices.

Marinella Research – Chandler, Arizona

Robotics Researcher

May 2024 – Dec 2024

- Lead multidisciplinary research projects in robotics, industrial automation, digital twins, and system engineering.
- Conduct feasibility studies and engineering consulting for industrial applications.
- Develop innovative solutions for commercial viability, emphasizing grant proposals and internal projects.
- Contribute to continuous improvement and technological advancement aligned with organizational goals.

Arizona State University – AZNext – Tempe/Mesa, Arizona

Teaching Assistant, Robotics in Microelectronics Manufacturing Training

Aug 2023 – Dec 2023

- Assisted in a cleanroom robotics workshop, introducing industrial robotics terminology, cleanroom requirements, vacuum requirements, robotic movements, and test requirements for robotics.

Naval Science & Technological Laboratory (NSTL) – Visakhapatnam, Andhra Pradesh

Robotics Intern

May 2022 – Aug 2022

- Developed Object Detection (YOLOv5) Robot with GPS navigation with vision-based collision avoidance.
- Worked on Python programable NVIDIA Jetson Nano JetBot.

Audi India – Visakhapatnam, Andhra Pradesh

Mechanical Technician · Internship

Mar 2017 – May 2017

- Diagnosed and fixed issues with mechanical components in Audi cars.
- Assembled and disassembled complex parts for repairs and troubleshoot malfunctions.

EDUCATION

Arizona State University – Mesa/Tempe, Arizona

GPA: 4.25

Doctor of Philosophy - PhD, Robotics and Autonomous Systems

Jan 2025 – Present

Master of Science in Engineering in Robotics and Autonomous Systems

Aug 2022 – May 2024

Bachelor of Science in Engineering in Robotics

Aug 2019 – May 2023

PUBLICATIONS

Journal Articles – Under Review

1. **Meesala, S.S.T.**, Rahmani, M., & Redkar, S. Kolmogorov–Arnold Networks Koopman Control of a 6 Degree of Freedom Robot Manipulator: Experimental Validation. *Control Engineering Practice*.
2. **Meesala, S.S.T.**, Rahmani, M., & Redkar, S. Auditable Smart Manufacturing: Kolmogorov–Arnold Oversight and Smart Contract Governance. *IEEE Transactions on Industrial Informatics*.

Conference Articles – Under Review

1. **Meesala, S.S.T.**, Rahmani, M., & Redkar, S. Multi-Agent Data-Driven Vehicle Control via Kolmogorov–Arnold Koopman Theory. *American Control Conference 2027*.
2. **Meesala, S.S.T.**, Rahmani, M., & Redkar, S. Data-Driven Autonomous Underwater Vehicle Control Using Bi-RRT, Koopman-KAN Learning, and Fractional Sliding Mode Control. *American Control Conference 2027*.

PROJECTS

SmartFollower & Tracker: TurtleBot 4 Autonomous Following System

Mar 2026 – May 2026

Developed a ROS 2 TurtleBot 4 system for ArUco-based object tracking, Kalman/Particle Filter prediction, SLAM mapping, and LiDAR-safe autonomous following using OAK-D vision and proportional control.

ArcFold: Inchworm-Inspired Foldable Robot (MuJoCo + Prototype)	Fall 2025
Modeled and validated a foldable rigid-link crawler in MuJoCo, incorporating compliance/actuator dynamics and sim-to-real gait tuning.	
AeroFusion: Autonomous BLIMP UAV (ROS 2 Sensor Fusion + Perception)	Spring 2025
Developed a ROS 2 indoor autonomy stack with IMU+barometer+camera fusion and YOLOv5-based target tracking.	
Robotics Dynamics and Force Control Package	April 2024
Developed a robotics package integrating manipulator dynamics, force control, and GUI.	
VCO2/VO2 Metabolic Monitor	Jan 2024 – Apr 2024
Designed and built a VCO2/VO2 metabolic monitor, integrating CAD design and embedded/software development with Bluetooth-enabled real-time data acquisition.	
Mechatronics Device: Portable/Lightweight Gait Analysis	Jan 2023 – May 2023
Developed compact and portable footwear for gait analysis to improve gait balance for patients at the Barrow Institute.	
Semi-Autonomous Twin Hoist System	Aug 2023 – May 2023
Designed two independent hoists to pick up and change copper rolls (700lbs) used for PCB manufacturing.	
Wearable Robotics with Embedded System Control	Jan 2022 – Apr 2022
Developed a sensor-driven embedded system with Wi-Fi control for actuator assistance in stroke rehabilitation.	
Industrial Workstation with Programmable Logic Controller	Jan 2022 – Apr 2022
Worked on a Rockwell PLC with industrial sensing technology, and industrial actuators, and developed Supervisory Control and Data Acquisition (SCADA) for the Human-Machine Interface (Machine: SMC HAS 200).	
3DoF Robot Arm	Aug 2021 – Dec 2021
Designed a Bluetooth Robot Arm with an inbuilt Forward/Inverse Kinematics calculator to reach the target location.	
Interactive Decoration/Animatronics with Embedded System Control	Aug 2021 – Dec 2021
Developed a Halloween decoration using embedded systems. In-depth research was done to design a functioning PCBs.	
SKILLS	
CAD Designing, Embedded Systems, Python, C, MATLAB, ROS, Linux, PLC Programming, Digital Twins, Aerial Robotics.	
EXTRACURRICULAR ACTIVITIES	
Winner – HGA Golf Tournament Category – A Sponsorship of INR 10,00,000 from Tusker Pharma	Runner Up – HGA Golf Tournament Category – A Participated in IGU-South Zone Golf Championship